

1. Lecture (advec-diff eq, PDE coordinates) 15 min
2. Exercises (20 min)
3. 5-10 min break
4. Lecture (method of lines, 15 min)
5. Exercise (Descrretization and work on code) rest of class

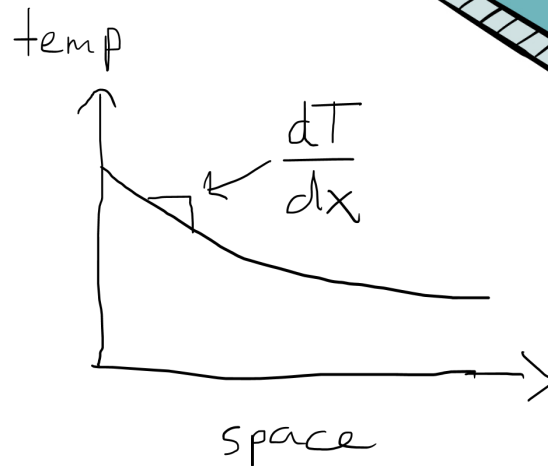
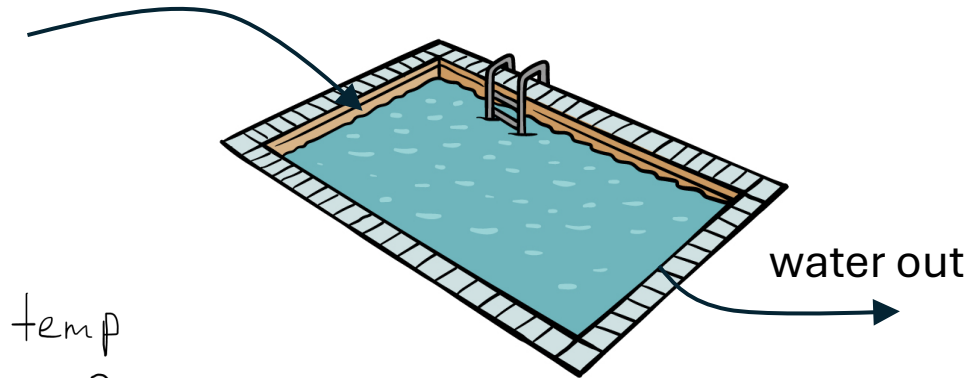
Models with spatial variation

Modelling Fall 2025

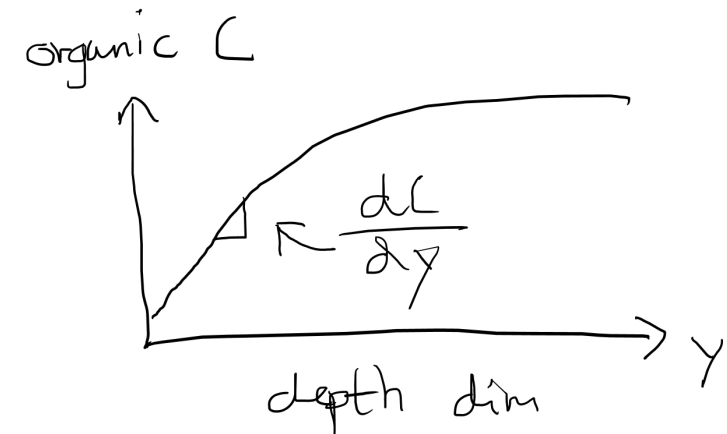
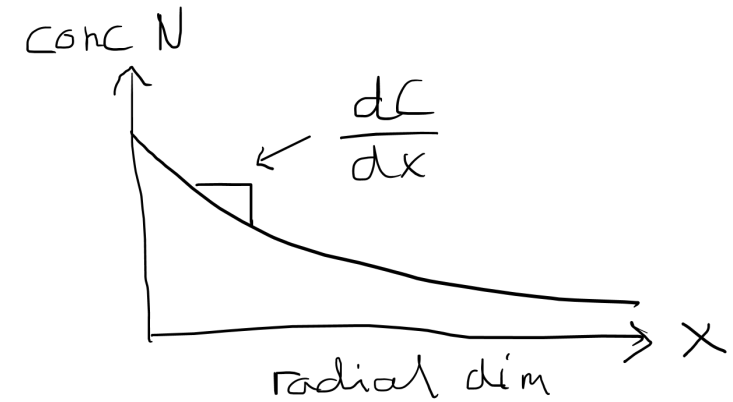
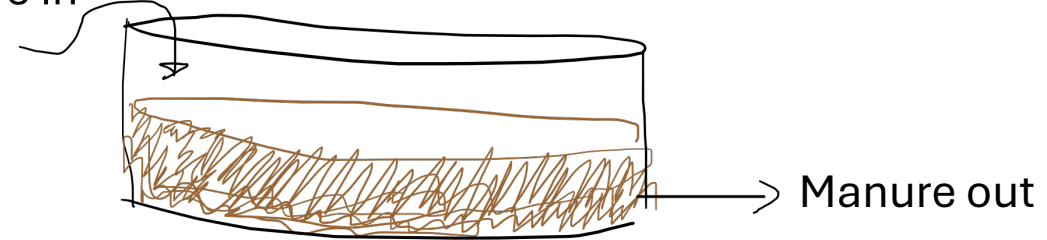
Frederik Dalby

Mini project examples

Warm water in



Manure in



From ODE to PDE

$$\text{acc} = \text{in} - \text{out} + \text{gen} - \text{con}$$

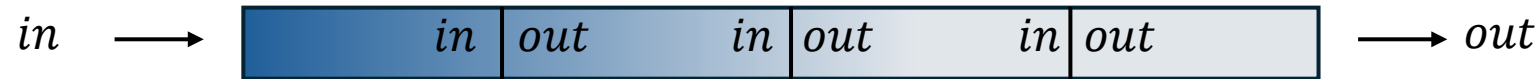
"In" and "out" terms can be thought of as sums of both diffusion and advection for the 0D problem

Lumped variables



$$\text{acc} = \text{in by advection} - \text{out by advection} + \text{in by diffusion} - \text{out by diffusion} + \text{gen} - \text{con}$$

ODE on each subdomain (distributed parameters)



Make subdomain infinitely small and x becomes a continuous independent variable



Then we can construct the governing a PDE!

$$\frac{\partial C_A}{\partial t} = -v \frac{\partial C_A}{\partial x} + D \frac{\partial^2 C_A}{\partial x^2} + \text{generation} - \text{consumption}$$

Change of C_A by advection (in by advection – out by advection)

constitutive

Change of C_A by diffusion (in by diffusion – out by diffusion)

constitutive

Ficks law of diffusion: $J = -D \frac{dC_A}{dx}$,

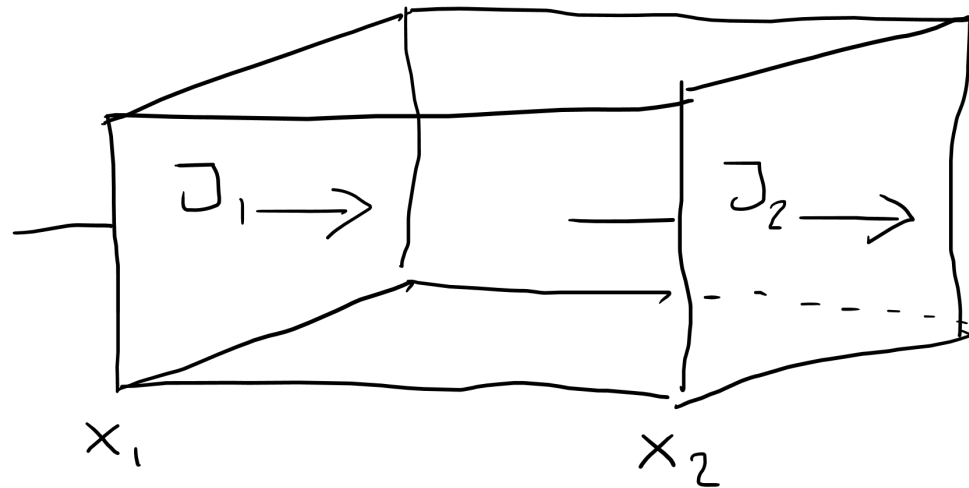
so why suddenly does $D \frac{\partial^2 C_a}{\partial x^2}$ represent diffusion?

We don't have a steady state anymore since

$\frac{dC_A}{dx}$ change with time **and space** in our 1D model!

Ficks law + law of conservation leads to Ficks second law ($\frac{\partial C_A}{\partial t} = D \frac{\partial^2 C_A}{\partial x^2}$)

Conservation law says: accumulation = flow in – flow out – generated - consumed



$$J = -D \cdot \frac{dc}{dx}$$



$$\frac{dc}{dt} = \frac{J_1 - J_2}{x_1 - x_2} = -\frac{\partial J}{\partial x} = -\frac{\partial \left(-D \frac{\partial c}{\partial x} \right)}{\partial x} = D \frac{\partial^2 c}{\partial x^2}$$

unit check : $\frac{g}{m^3 s} = \frac{\frac{g}{m^2 s} - \frac{g}{m^2 s}}{m - m} = \frac{g}{m^3 s}$

See uploaded notes on ficks law

$$\frac{\partial C_A}{\partial t} = -v \frac{\partial C_A}{\partial x} + D \frac{\partial^2 C_A}{\partial x^2} + \text{gen} - \text{cons}$$

Is the advection-diffusion-reaction equation → common in chemical/biological engineering

What if we had spatial variation in more dimensions??

Governing equation for mass

$$\frac{\partial C_A}{\partial t} = -\nabla \cdot (vC_A) + D_{AB} \nabla^2 C_A + R_A$$

advection diffusion reaction
 (generation)

$$\nabla \cdot (vC_A) = v_x \frac{\partial C_A}{\partial x} + v_y \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z}$$

Governing equation for energy

$$\frac{\partial T}{\partial t} = -\nabla \cdot (vT) + \alpha \nabla^2 T + S$$

Thermal advection Thermal diffusion Heat source (generation)

$$\alpha = \frac{k}{\rho C_p}$$

Thermal diffusivity Conductivity coefficient Heat capacity Density

Coordinates of PDE equations

Rasmuson Appendix A

Rectangular (cartesian)

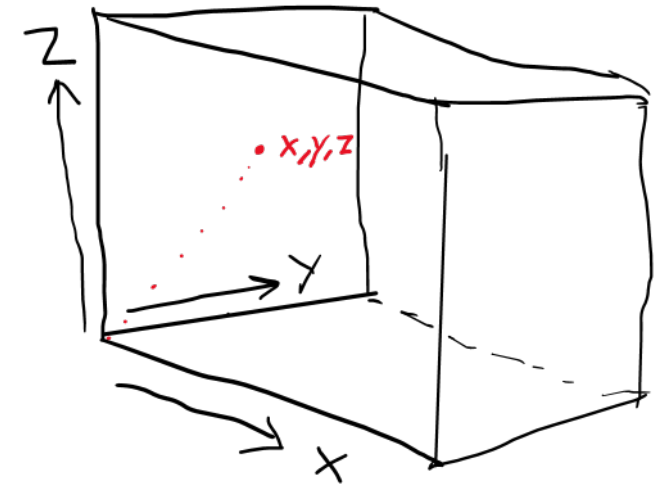
Governing equation for mass

$$\frac{\partial C_A}{\partial t} + v_x \frac{\partial C_A}{\partial x} + v_y \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z} = D_{AB} \left(\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2} \right) + R_A.$$

Governing equation for energy

$$\rho c_p \left(\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) + S.$$

$$\alpha = \frac{k}{\rho c_p}$$



Cylindrical coordinates

Governing equation for mass

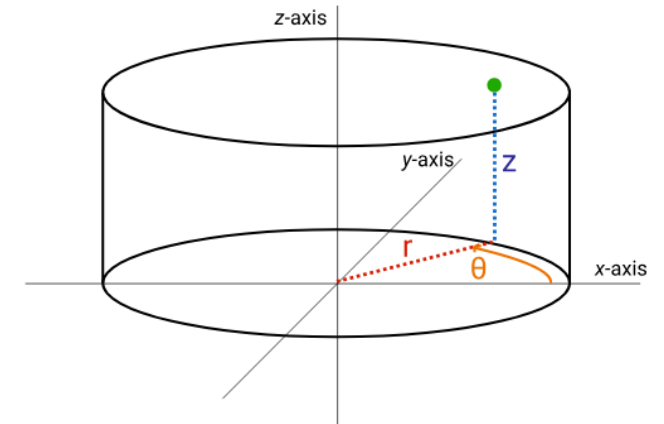
$$\frac{\partial C_A}{\partial t} + v_r \frac{\partial C_A}{\partial r} + v_\theta \frac{1}{r} \frac{\partial C_A}{\partial \theta} + v_z \frac{\partial C_A}{\partial z} = D_{AB} \left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial C_A}{\partial r} + \frac{1}{r^2} \frac{\partial^2 C_A}{\partial \theta^2} + \frac{\partial^2 C_A}{\partial z^2} \right) + R_A.$$

Governing equation for energy

$$\rho c_p \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + v_\theta \frac{1}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = k \left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right) + S.$$

Governing equation for mass

$$\frac{\partial C_A}{\partial t} + v_r \frac{\partial C_A}{\partial r} + \cancel{v_\theta \frac{1}{r} \frac{\partial C_A}{\partial \theta}} + v_z \frac{\partial C_A}{\partial z} = D_{AB} \left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial C_A}{\partial r} + \cancel{\frac{1}{r^2} \frac{\partial^2 C_A}{\partial \theta^2}} + \frac{\partial^2 C_A}{\partial z^2} \right) + R_A.$$



if symmetry
for all θ

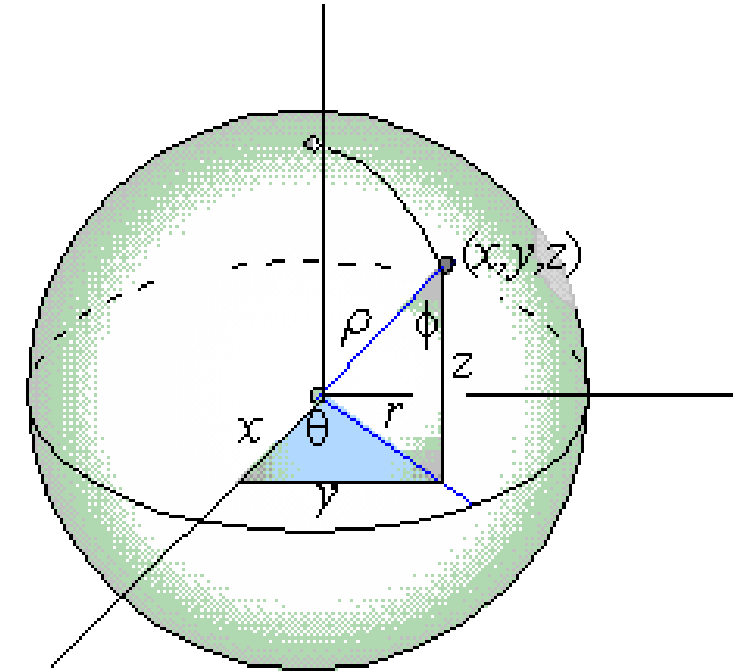
Spherical coordinates

Governing equation for mass

$$\begin{aligned} \frac{\partial C_A}{\partial t} + v_r \frac{\partial C_A}{\partial r} + v_\theta \frac{1}{r} \frac{\partial C_A}{\partial \theta} + v_\phi \frac{1}{r \sin \theta} \frac{\partial C_A}{\partial \phi} \\ = D_{AB} \left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial C_A}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial C_A}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 C_A}{\partial \phi^2} \right) + R_A. \end{aligned}$$

Governing equation for energy

$$\begin{aligned} \rho c_p \left(\frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + v_\theta \frac{1}{r} \frac{\partial T}{\partial \theta} + v_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \right) \\ = k \left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial T}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial T}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} \right) + S. \end{aligned}$$



Governing equation for mass

$$\begin{aligned} \frac{\partial C_A}{\partial t} + v_r \frac{\partial C_A}{\partial r} + \cancel{v_\theta \frac{1}{r} \frac{\partial C_A}{\partial \theta}} + \cancel{v_\phi \frac{1}{r \sin \theta} \frac{\partial C_A}{\partial \phi}} \\ = D_{AB} \left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial C_A}{\partial r} + \cancel{\frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial C_A}{\partial \theta}} + \cancel{\frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 C_A}{\partial \phi^2}} \right) + R_A. \end{aligned}$$

for symmetry for all θ and ϕ